

number of counts. For a 6000-count multimeter with maximum DC voltage range of 600V, resolution will be $600/6000 = 0.1V$.

Accuracy. Accuracies of all the measurement features are mentioned in the datasheet. First, check basic DC volts accuracy. Accuracy is given in percentage (like 0.5 per cent). For regular electronics work, you need not use a multimeter with a very high accuracy like 0.05 per cent. You can do reasonably well with just 0.5 per cent accuracy.

Accuracy of other measurement functions is also important and should be very close to basic DC volts accuracy. Current measurement accuracy will not be so much high as a shunt resistor is used to measure current.

Input impedance. Ensure that the multimeter you choose has an input impedance as high as 10 megaohms. When you measure voltage across a component, you are effectively putting the input resistance of the multimeter in parallel to the resistance of the component. Therefore a small input impedance will change the values you are trying to measure.

True RMS. Two types of multimeters are available: average-responding and true RMS (root mean square). True RMS is measurement of AC voltage or current that reflects the amount of power dissipated by a resistive load driven by the equivalent DC value. An average-responding AC multimeter is calibrated to read the same as a true-RMS meter for sine wave inputs only. For other waveform shapes, an average-responding meter will give substantial errors.

Updating speed. General-purpose multimeters are available with display update rate of five per second. The

higher the update rate, the better.

Battery life. A good battery life will make your life easier with electronics. Some advanced multimeters are quite power-hungry. Buy them only if you really require their advanced features. It is recommended to have a general-purpose multimeter with a battery life of 300 to 500 hours.

Measurement convenience

Below-mentioned features can make measurements very convenient, which means a lot while struggling with electronics circuits:

Auto range with manual override.

In a manual-range multimeter you have to set the range before taking a measurement, which is very tedious. Go for an auto-range multimeter if you are serious about your electronics project. Ensure manual override function too as you will need it sometimes.

Automatic touch hold. This feature is very handy when both your hands are engaged in taking the measurement and you cannot look at the display for the reading. With this feature you can press the hold button and take measurement without looking at the display. The measured value will stay there until you reset it. Make sure that this feature is automatic, which means that you don't have to press any button to freeze the reading because you will not have a spare hand to press the hold button.

Relative measurement. You can use this feature to analyse relative changes in the measured value. You can also use it to compensate for losses in the probes: Short both the probes and keep the relative measurement button pressed until it shows '0.' This is very useful in resistance and voltage measurements.

Min/max mode. Hook up the multimeter to a circuit and use min/max mode to get the minimum and maximum of the values it recorded during this time period. This feature makes measurement of minimum and maximum values very easy as you don't have to continuously look at the reading display.

Duty-cycle measurement. This is not really an essential feature in a multimeter as you will never measure the duty cycle of waveforms using a multimeter. There are other equipment for such measurements, such as an oscilloscope.

Display size, contrast and backlight. It is always better to go for a bigger display so that the readings can be read easily from a distance. Also, the display contrast should be high. Cheap Chinese multimeters have a very poor display contrast. Backlight is useful for taking measurements in dark.

Capacitance, temperature and frequency measurements. Capacitance measurement with a multimeter is not very accurate and you will need an LCR meter for that. However, a multimeter with capacitance measurement accuracy of up to 2 per cent is reasonably good. Temperature measurement is a highly useful feature that will reduce the number of test equipment from your table. Frequency measurement is not that useful but if you are getting it, look for the highest possible range.

Low-pass filter. Low-pass filter in a multimeter helps to block unwanted voltages above 1 kHz when measuring AC voltage or AC frequency. It can improve measurement performance on composite sine waves that are typically generated by inverters and variable-frequency motor drives.

Kelemon
step into the new world of 3D experiences

New TRION Series V6X3D3HD ₹23,000/-
New TRION Series V6X3D47HD ₹48,500/-

ktab TABLET PCs manufacturer
Learning solution For 21st CENTURY

Designs inc
Franchise Enquiry Solicited for Android Development Training
Trade Enquiry admin@ktab.in 099943 56789 098941 83516
www.ktab.in www.kelemon.in www.designsinc.in

Dealers & Distributors Wanted